

TRANSMISSION MEMORANDUM & FORMAL PITCH EMAIL

Date: September 24, 2026

To: Muhammad Rafay Shaikh

From: Muhammad Khalil (Department of Public Administration, University of Karachi)

Subject: Formal Proposal: AI-Assisted Outpatient Flow Optimization (Smart Patient Flow)

Attachment: Executive Project Proposal - Smart Patient Flow.pdf

Email Body

Dear Concern,

Prolonged outpatient waiting times remain a persistent operational challenge across tertiary healthcare facilities, driven largely by sequential visit processing, manual triage bottlenecks, and administrative burden on clinical staff.

To address this, I am formally submitting the project proposal for **Smart Patient Flow**—an intelligent operational architecture engineered to compress door-to-doctor transit times while strictly preserving complete clinical autonomy for hospital physicians.

Key Operational & Technical Highlights

- **Parallel Diagnostic Processing:** Medical committee-approved standing orders run concurrently during waiting intervals, ensuring preliminary lab results are accessible on-screen prior to doctor consultation.
- **Deterministic, Doctor-Governed Triage Engine:** Patient sorting is driven by explicit clinical rule-sets drafted and signed by departmental leadership—eliminating unverified, black-box diagnostic algorithms.
- **Preserved Clinical Authority:** Attending physicians maintain sole authority over diagnosis, clinical plans, and prescription signing.
- **Unrestricted Floor Override:** Floor nurses and clinicians retain immediate, uninhibited capability to manually reassign or override queue positions at any point.
- **Non-Disruptive HMIS Interoperability:** Functions as an operational integration layer connecting with existing registration, laboratory, and electronic records systems via HL7/FHIR protocols without core system replacement.
- **Evidence-Based Phased Deployment:** Execution initiates with an empirical two-week baseline time study, transitioning through an offline shadow-mode trial before live departmental activation.

Financial & Resource Summary (Single Pilot Department)

- **Deployment Scope:** 12-month pilot covering Phases 0 through 3 (Discovery, Clinical Design, Core Build, and Supervised Beta).
- **Technical Effort (Line A):** **PKR 18,200,000** (104 person-months across specialized engineering, architecture, process analysis, and QA roles).
- **Hardware & Infrastructure (Lines B–D):** **PKR 3,300,000** (self-service check-in kiosks, digital waiting-room displays, clinical tablets, cloud/telephony gateways, and independent penetration testing).
- **Contingency Reserve (Line E):** **PKR 4,300,000** (20% standard operational allocation).
- **Total Estimated Pilot Investment:** **PKR 25,800,000**
- **Annual Post-Go-Live Support:** Projected at 15–20% of build cost (~PKR 3.8M–5.1M/year) for routine maintenance, security testing, and regular clinical audits.

The complete executive proposal document—including phase deliverables, safety governance, and itemized budgetary schedules—is attached for your review.

I would welcome the opportunity to schedule a brief discussion at your convenience to review the implementation roadmap and explore running a baseline study within your outpatient facility.

Sincerely,

Muhammad Khalil

Department of Public Administration

University of Karachi